

Qatar-2b

R-0660

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_260409 · created 2026-07-27T17:39:27+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461139.8695 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.065
Transit depth (Rp/Rs)^2	5.19 +/- 2.95 [%]
Orbital Inclination (inc)	85.25 +/- 1.72
Airmass coefficient 1 (a1)	1.31 +/- 0.65
Airmass coefficient 2 (a2)	-0.18 +/- 0.68
Scatter in the residuals of the lightcurve fit is	46.72 %
Best Comparison Star	#3 - [473, 88]
Optimal Aperture	1.7
Optimal Annulus	4.01
Transit Duration (day)	0.071 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.228 $\pm$ 0.065 vs 0.16327 (deviation 0.87 σ)
Duration fit vs published	0.071 d vs 0.0767 d (deviation 0.64 σ)
Mid-time O-C	-4.7 min
Depth SNR	3.1
Residual scatter	46.72 %

Light curve

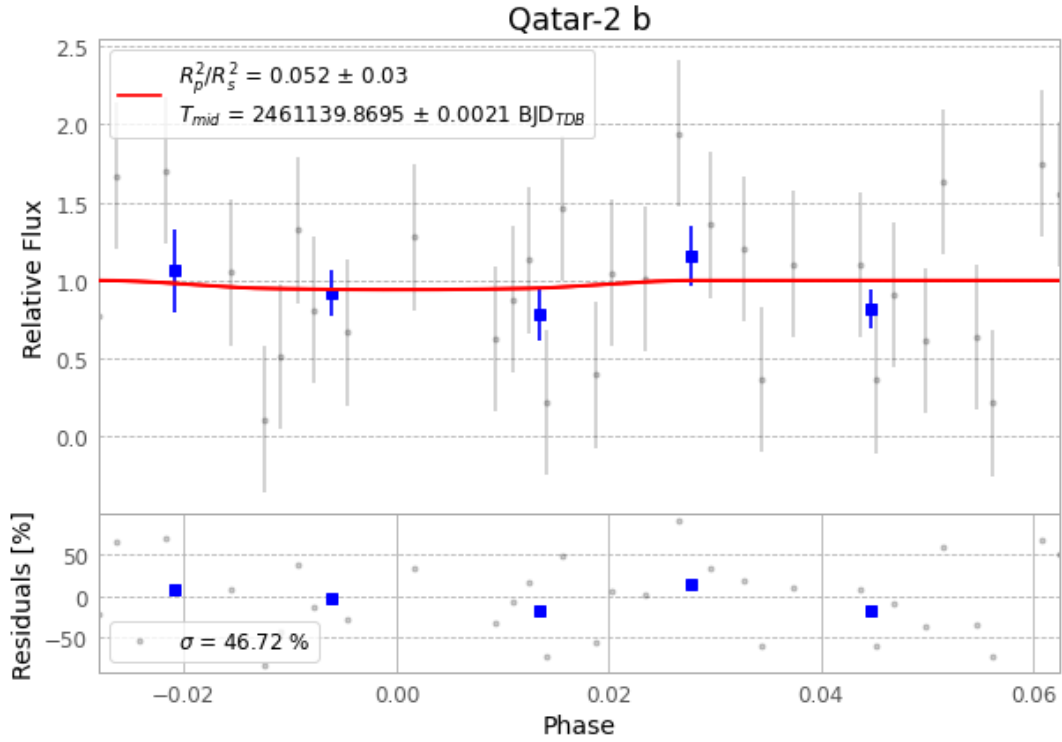


Figure 1 — FinalLightCurve_Qatar-2 b_2026-04-09.png (SHA-256 556aff7540cb679e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [164, 450], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2ebbaa44dd778d99...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

60 FITS frames (39 MB), Qatar-2260409074215.FITS → Qatar-2260409104215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-260409.log (5d53d27d05cb...), FinalLightCurve_Qatar-2 b_2026-04-09.png (556aff7540cb...), FinalLightCurve_Qatar-2 b_2026-04-09.csv (e93a8d281d03...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:39Z.